L&P Intelligence Platform

Business Objectives and Reference Architecture

1 · BUSINESS OBJECTIVES

Five business objectives and the platform capabilities each requires

PRIMARY

Stronger proposals

- Accurate, complete, compliant proposals for each RFP
- Know what was strong, what failed and what to change
- Staffing and price plans leadership can approve

Document intake · Compliance matrix · Pricing engine · Win/loss history · Price-to-win

GROWTH

Find and convert business

- Which buyers to pursue, when, and with what advantage
- A qualified next-contact list with “why now”
- Which sources and messages win customers

Market analysis · Recompete calendar · Lead scoring · Pwin · Outreach tracking

PROTECTION

Protect revenue and relationships

- Authorization gaps and renewals flagged in time
- Contract economics and billing exceptions explained
- Client issues owned before renewal

Contract records · QuickBooks + Belfry exports · Alerts · Client reviews

READINESS

Strengthen company capability

- Verified past performance ready to reuse
- Credential and eligibility gaps visible
- Partner, expansion and training plans

Evidence library · Credential tracking · Expansion tracking

SHARED

Reusable organizational intelligence

- Every bid adds knowledge for the next
- ChatGPT and Claude work from the same records
- Less repeated research and re-entry

One source of truth · AI agent access · Automation

ALSO REQUESTED — Pwin (probability of win) · Price-to-win · Market analysis · Lead scoring · AI automation for speed and accuracy

1 · Collect	2 · Verify	3 · Analyze	4 · Decide	5 · Produce	6 · Learn
RFPs, our bids, competitor awards, market data	AI extracts, staff confirm, every fact sourced	Comparisons, Pwin, price-to-win, leads	Bid or not, staffing, price	Proposal, forms, price sheets	Outcome and feedback feed the next bid

Layered architecture from data ingestion to user access, across three build options

Sources — RFPs, past bids, public awards, QuickBooks, Belfry exports → **Ingest** — Connectors + workers (Cloud Run jobs, worker pools) → **Store** — Database, files, search (Supabase) → **Analyze** — Warehouse + AI (Google Cloud) → **Back end** — API, sign-in, permissions (Vercel) → **Access** — Claude Enterprise / ChatGPT first, plus staff web app

Use · Intelligence · Data · Operations — Included · Partial · + Added by this option

	1 · Intelligence platform Smallest real starting point	2 · Enterprise data & AI Pwin, price-to-win, leads, market	3 · Full operating platform Everything in the Reference sheet
SOURCES IN · Cloud Run **Connectors** Cloud Run jobs · Cloud Scheduler	Texas ESBD + local portals, Drive, QuickBooks, Belfry exports	+ USAspending, SAM.gov, wage determinations, bid tabs (records requests)	+ Apollo / ZoomInfo, Gmail, Google reviews (approved access)
Workers Python · Cloud Run jobs + worker pools	OCR, document parsing, portal crawling, document output	+ More portals, continuous crawling	+ Always-on crawling and enrichment
DATA · Supabase **Database** PostgreSQL · system of record	**Bids, requirements, prices, staffing, contracts, competitors, outcomes**	+ History tables, staging project, point-in-time recovery	+ Leads, outreach, client reviews, credentials
File storage Supabase Storage	Original RFPs, proposals, contracts; each version kept as a new file	+ Public awards and bid tabulations	Same as option 2
Search Full-text + pgvector, inside PostgreSQL	Keyword (full-text) + meaning (embeddings) search over verified text	Same as option 1	Same as option 1
ANALYTICS + AI · Google Cloud **Data warehouse** BigQuery · Dataform	Not yet: reports run on the database	+ Multi-year history; Pwin and price models (BigQuery ML or Python)	+ Marketing, client and expansion analytics
AI services Gemini · Agent Platform (formerly Vertex AI)	Extraction, drafting, embeddings for search	+ Model insights on scores	+ Agent automation at volume
BACK END · Vercel **Backend / API** Next.js API · auth · Vercel Workflows	Staffing and pricing engine, approvals	+ Pwin scorecard, price-to-win, lead scoring	+ Outreach sequences, records-request follow-up
ACCESS (FRONT END) · Vercel **AI agents (primary)** Claude Enterprise + ChatGPT via MCP	Each person signs in; agents read, search and update with that person's permissions	Same as option 1	+ Scheduled agents (Claude routines or API agents)
Staff web app Next.js · review, approvals, dashboards	Bid, pricing and contract workspace	+ Market and pipeline dashboards	+ Sales, client and executive views
ACROSS ALL LAYERS **Operations** GitHub CI/CD · Sentry · Cloud Monitoring	Backups, monitoring, alerts	+ Staging and production	+ Agent cost and safety limits

Why the database comes first: it is the system of record; only the back end and workers read or write it. The current 87-tab sheet is about 2.35M tokens, more than twice the 1M-token windows of current Claude, GPT and Gemini models, and AI accuracy drops as input grows. A database lets an agent query only the records it needs.